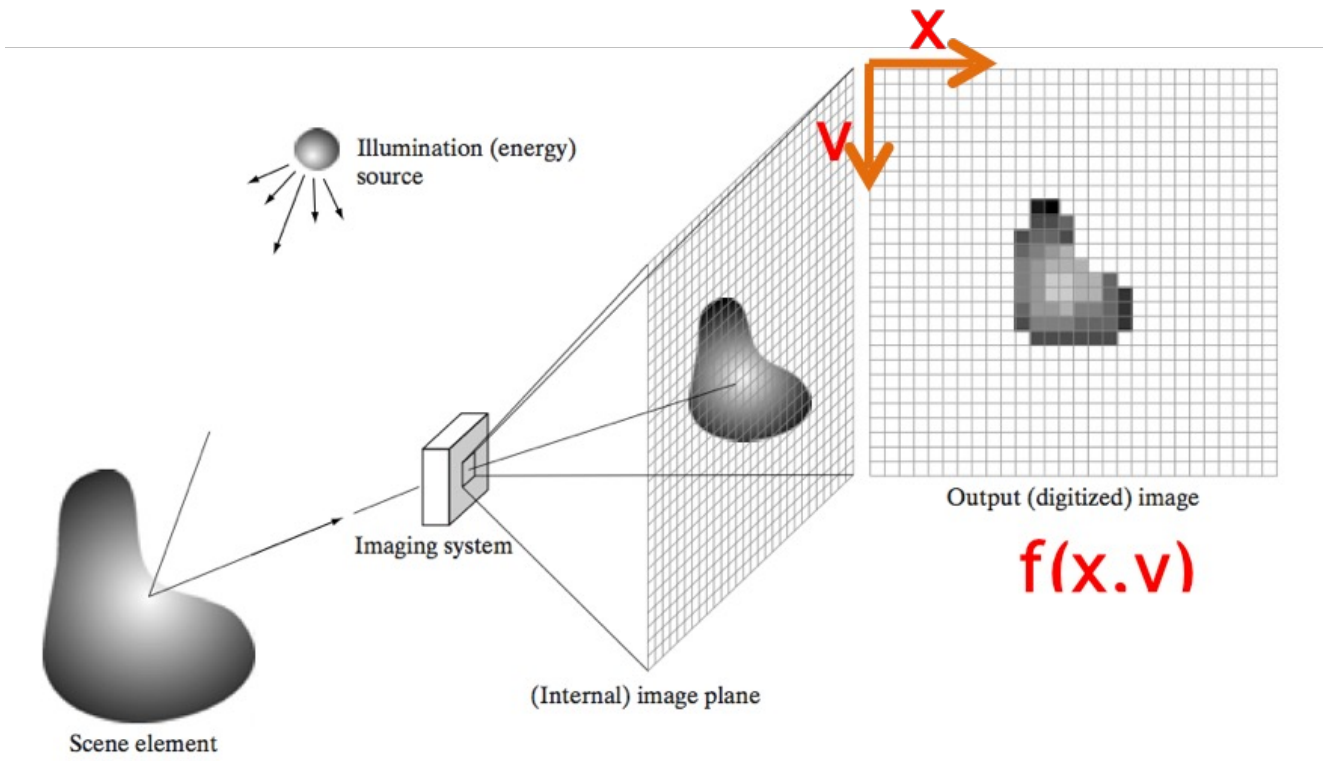
Camera Models

Avinash Sharma

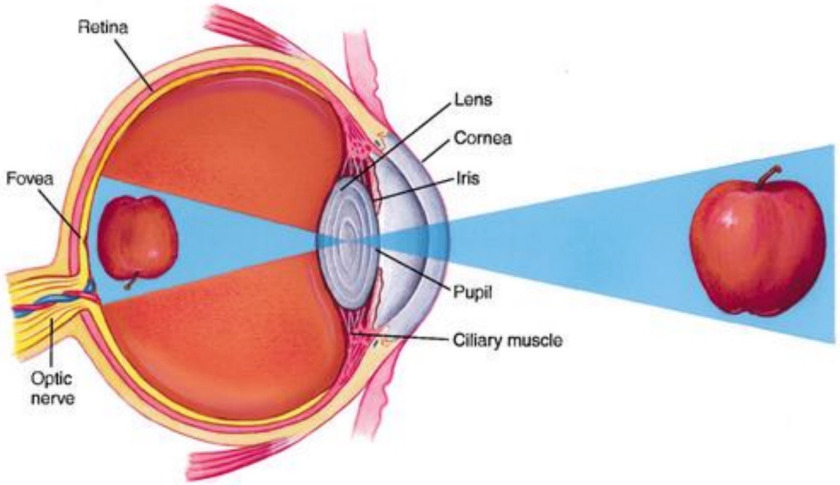
CSE Department, IIT Jodhpur

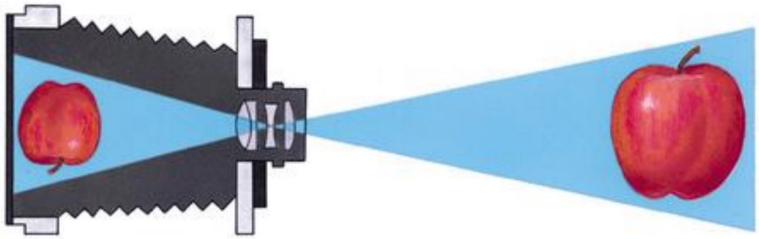
Image courtesy: https://www.etsy.com/in-en/listing/1607436850/third-eye-activation-spiritual-awakening

Image Formation

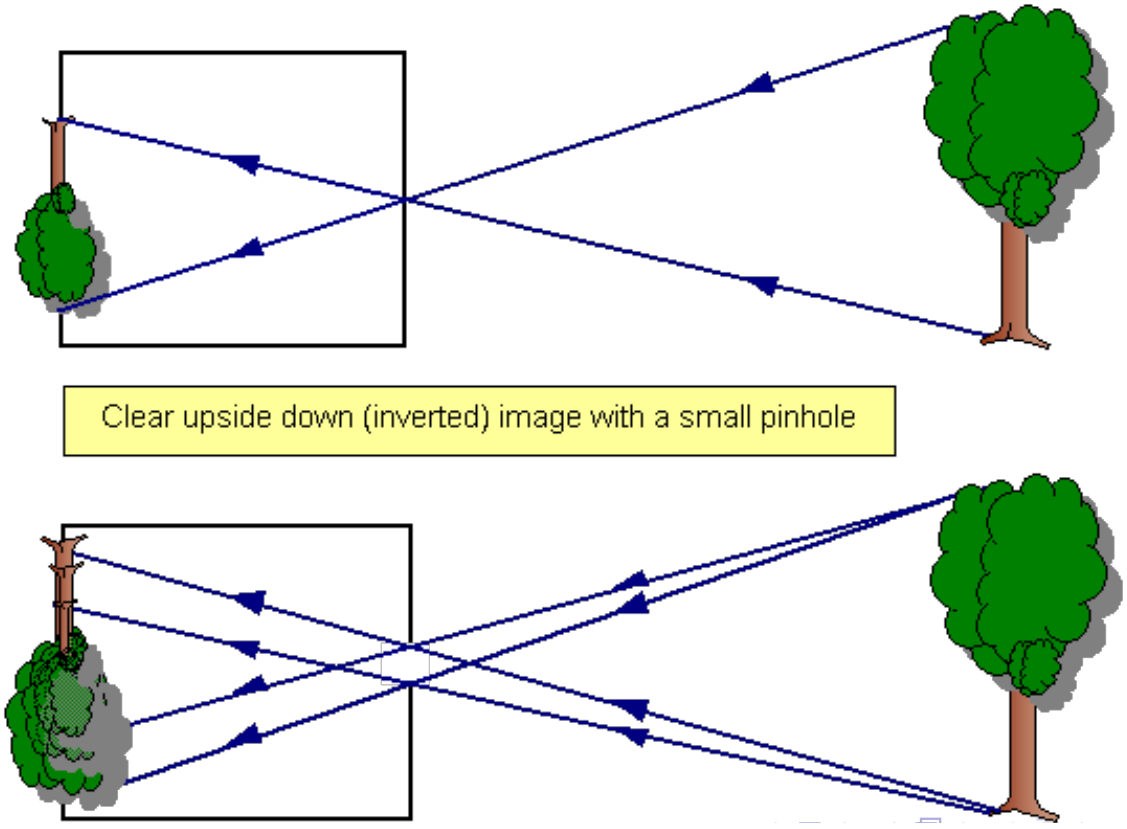


Imitating Human Vision

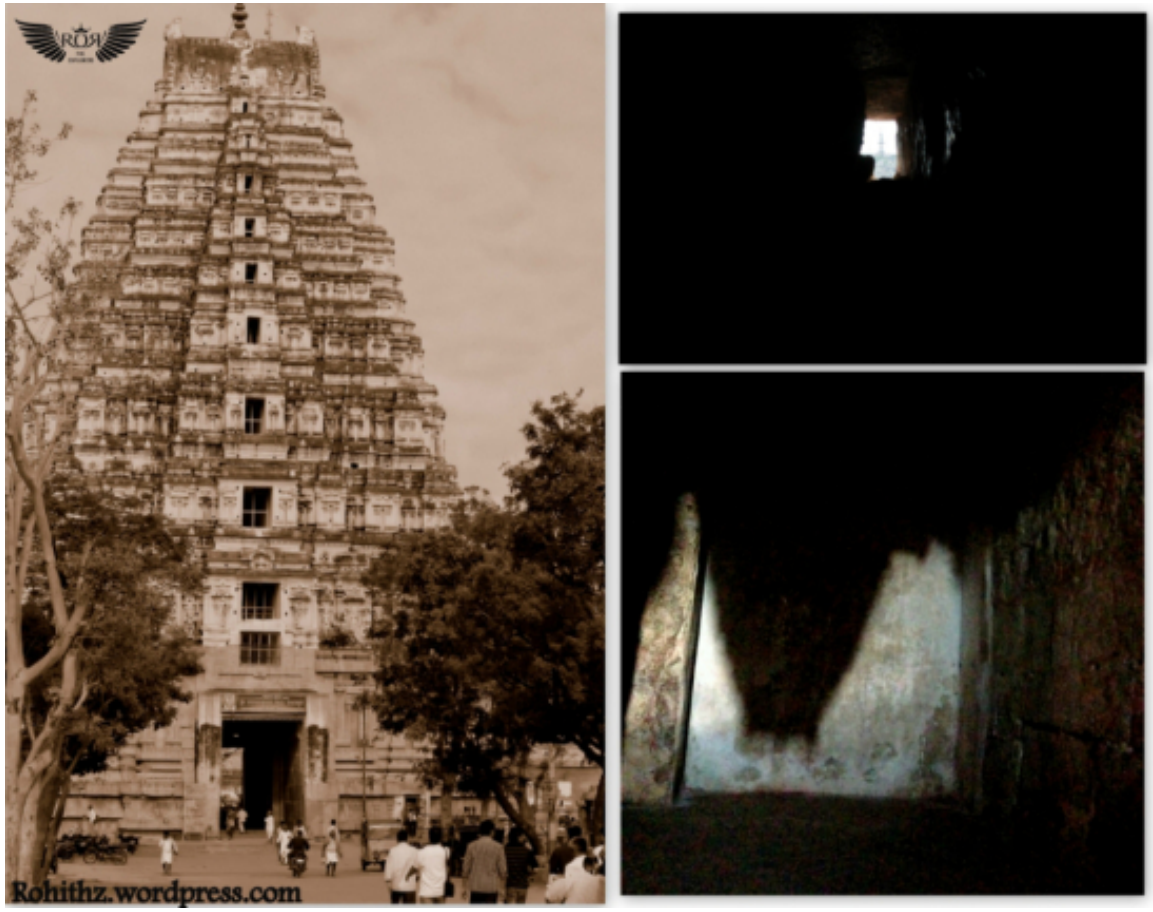




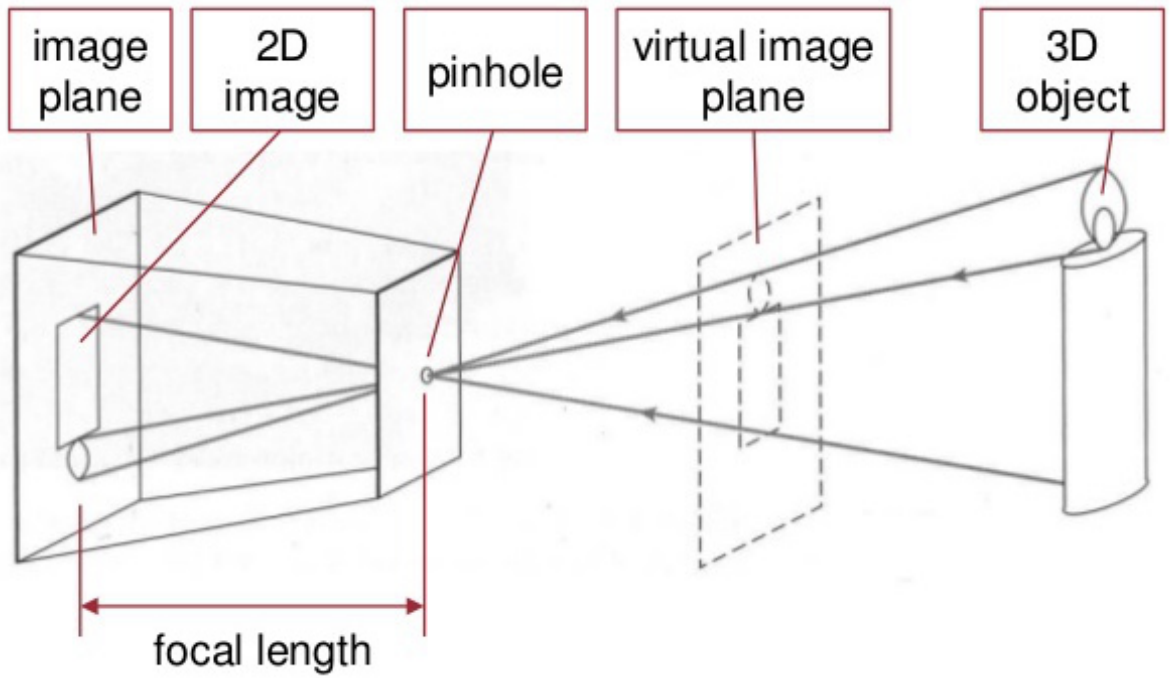
Pin-hole Camera Model



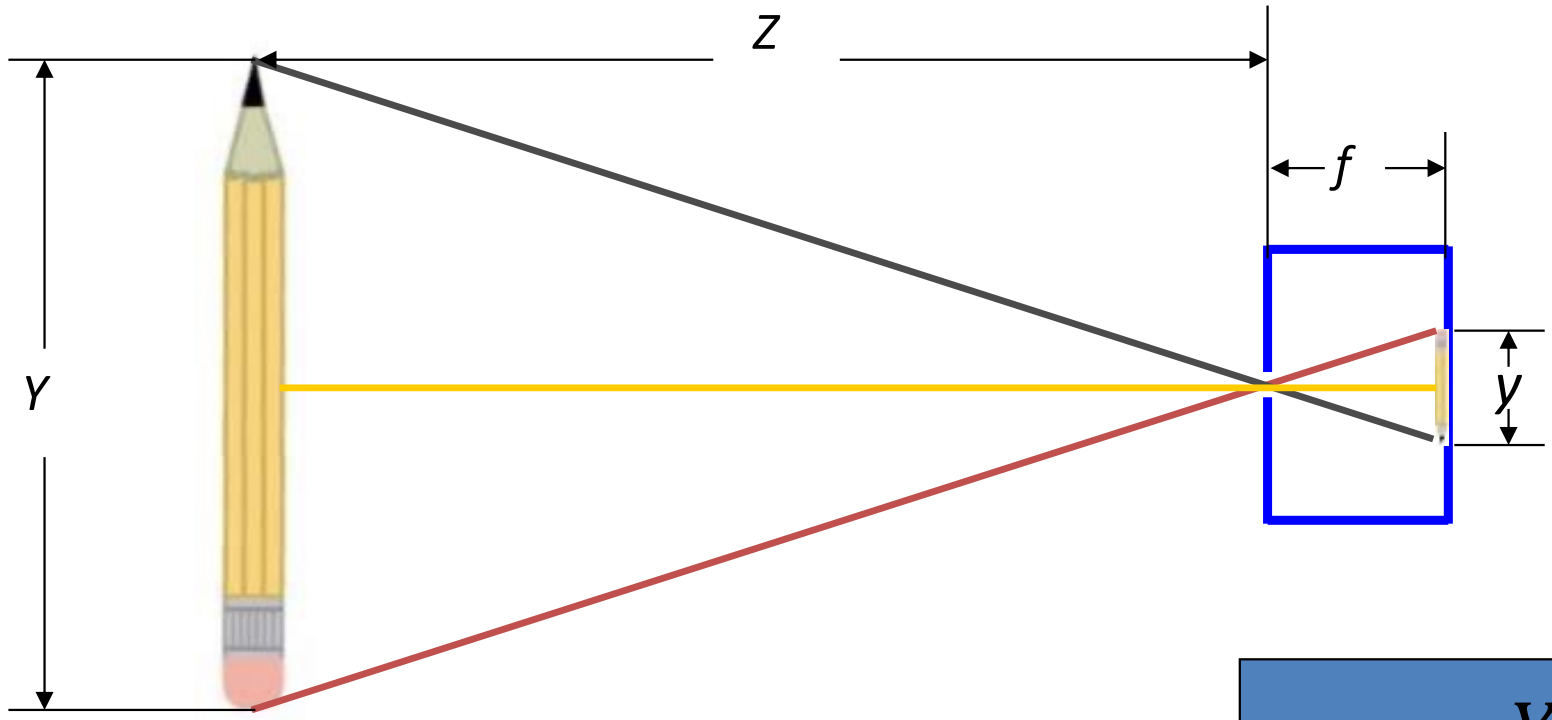
Pin-hole Camera Model



Pin-hole Camera Model



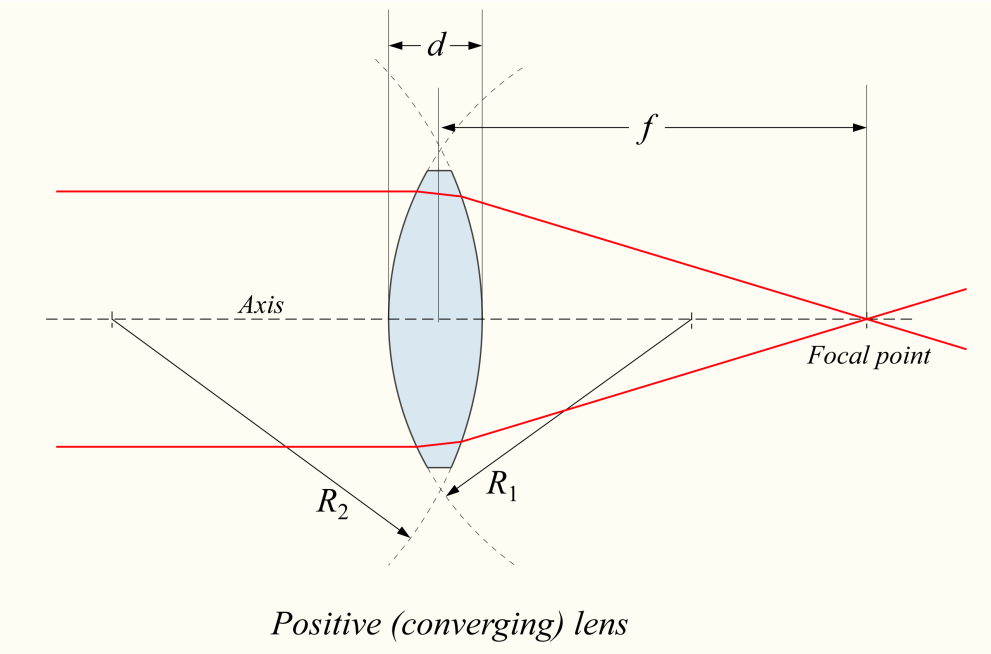
Pin-hole Camera Model



$$y = f\frac{Y}{Z}$$

Camera with Lens

In optics, a thin lens is a lens with a thickness that is negligible compared to the radii of curvature of the lens surfaces.



The focal length, f, of a lens in air is given by the Lensmaker's equation:

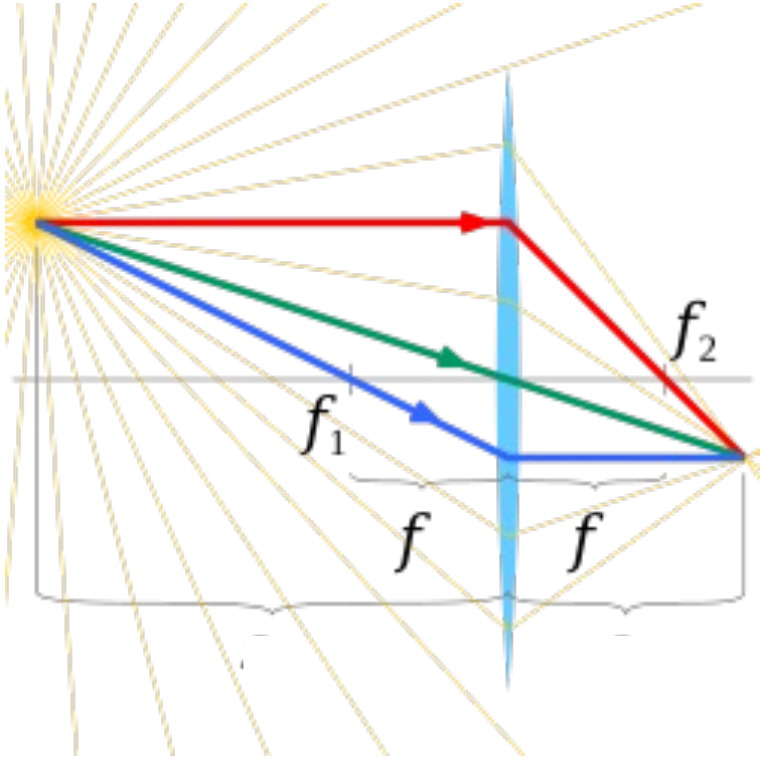
$$\frac{1}{f} = (n-1)\left[\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n-1)d}{nR_1R_2}\right],$$

where n is the index of refraction of the lens material, R1 and R2 are the radii of curvature of the two surfaces, and d is the thickness of the lens.

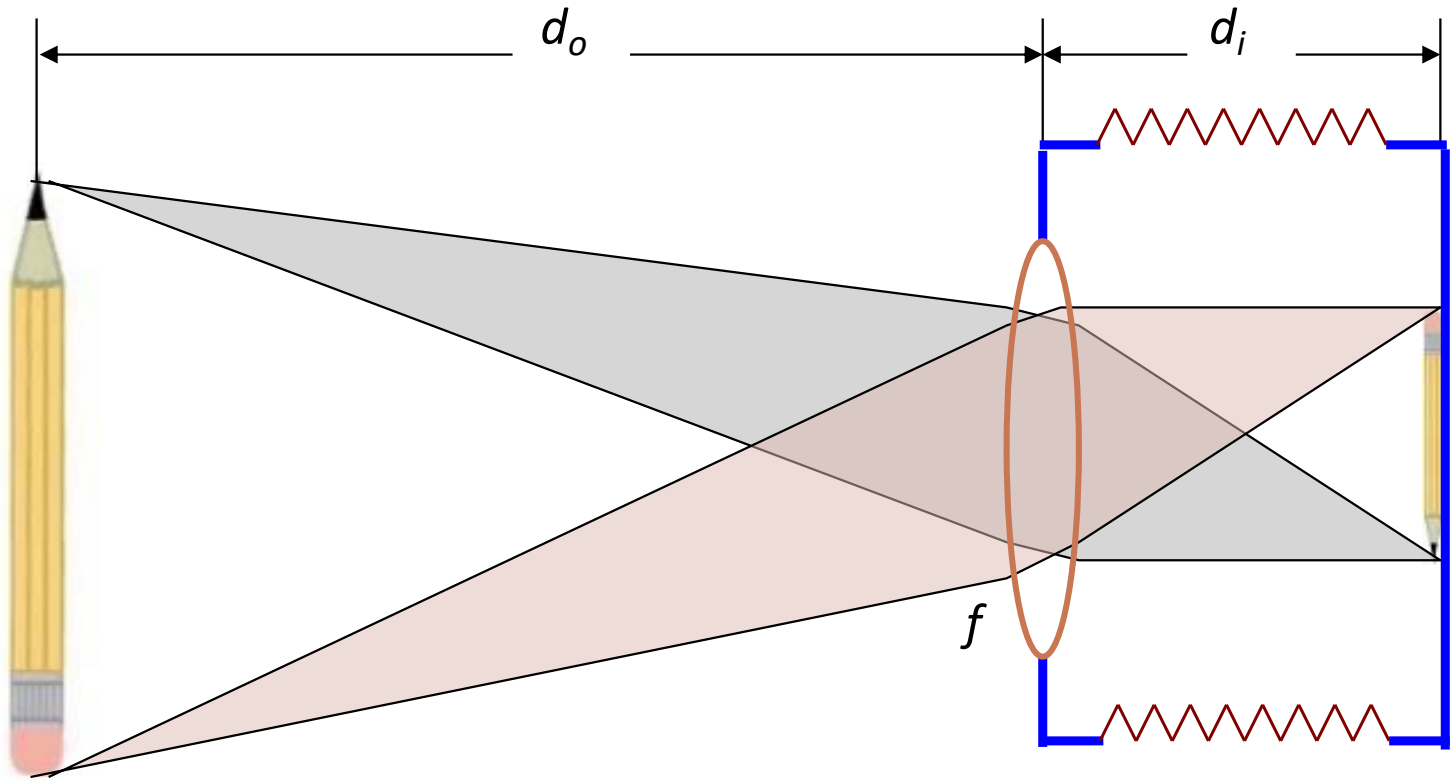
Camera with Lens

Certain rays follow simple rules when passing through a thin lens, in the paraxial ray approximation:

- Any ray that enters parallel to the axis on one side of the lens proceeds towards the focal point on the other side (red in figure).
- Any ray that arrives at the lens after passing through the focal point on the front side, comes out parallel to the axis on the other side (blue).
- Any ray that passes through the center of the lens will not change its direction (green).



Camera with Lens



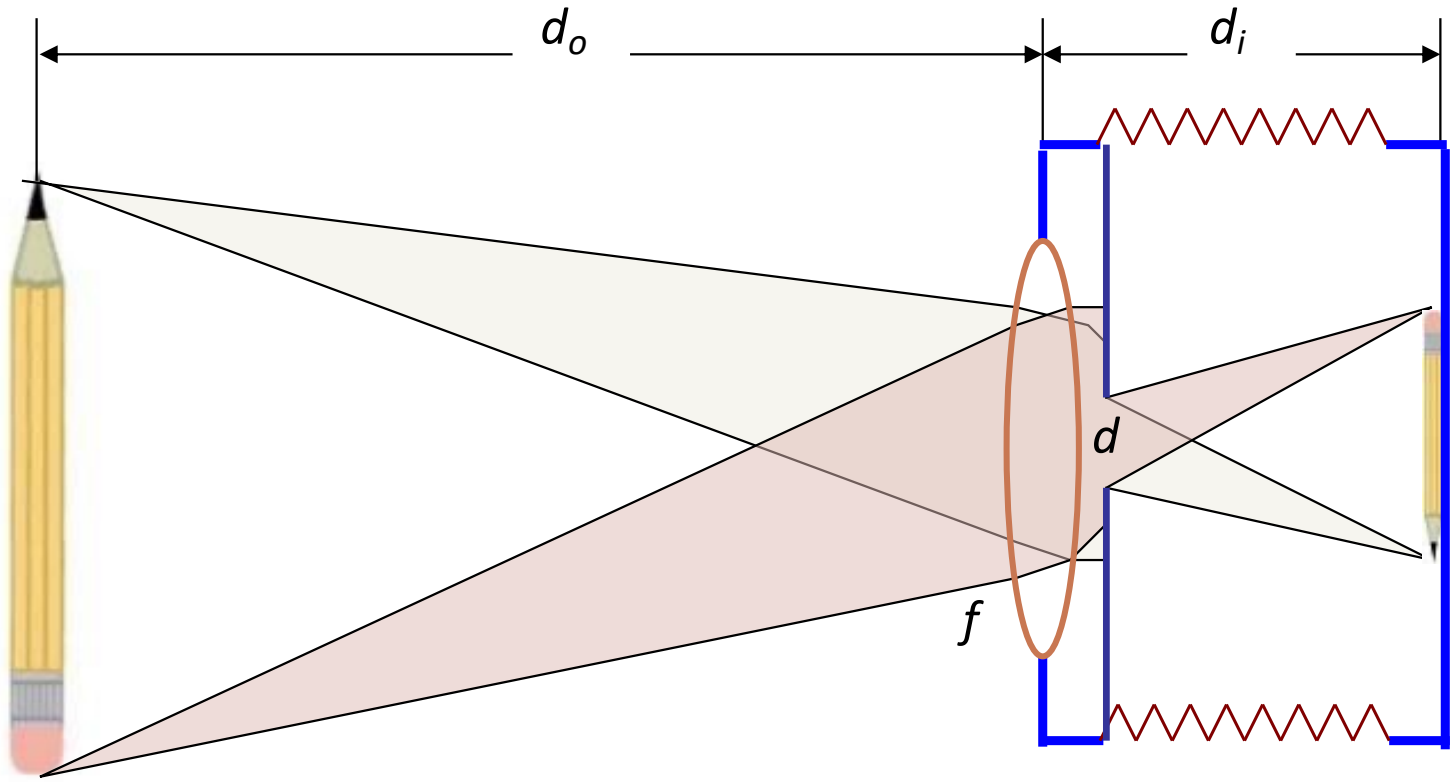
Thin lens equation: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

$$d_i = f\frac{d_o}{(d_o - f)}$$

https://www.youtube.com/watch?v=x9F0AQCGkbc

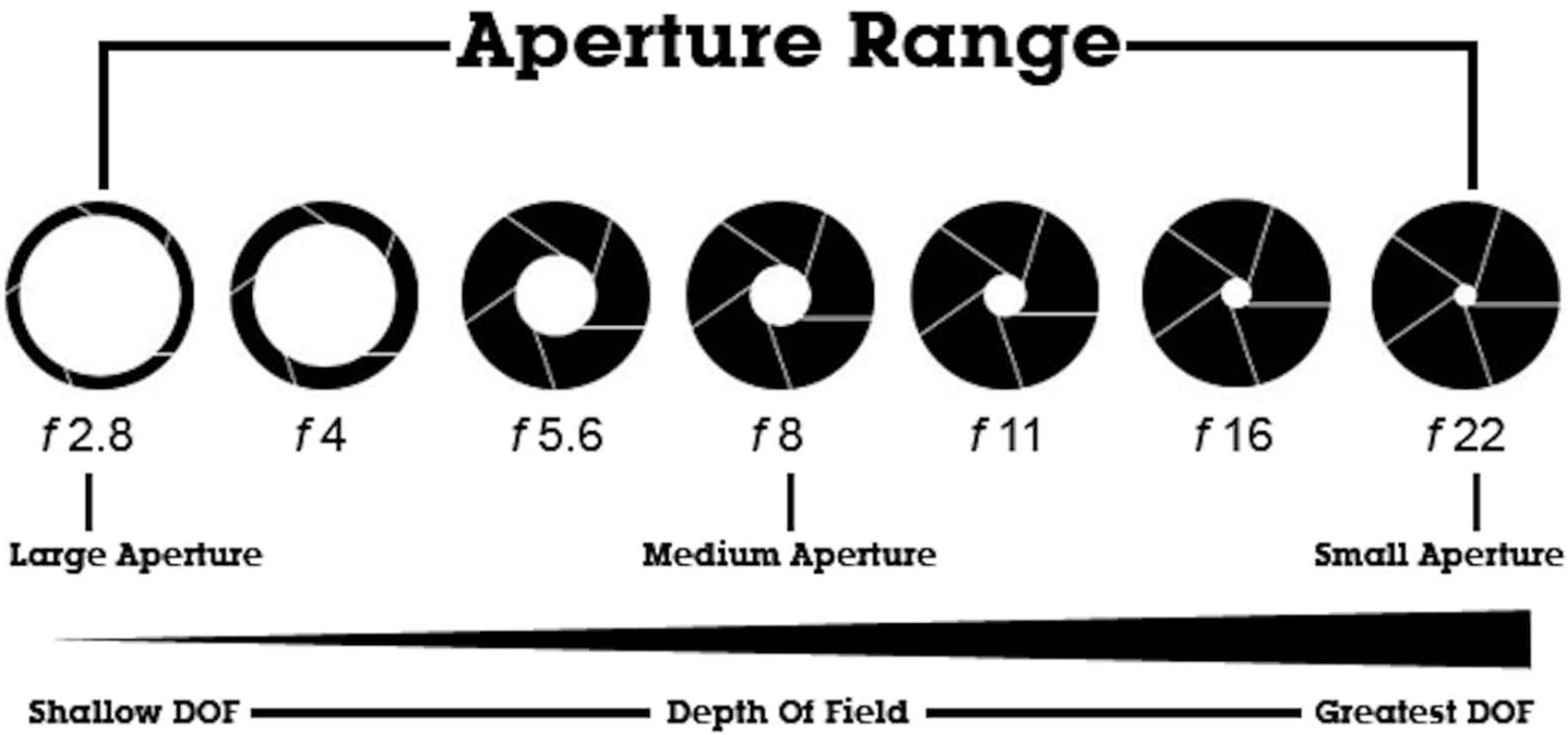
Focus and Depth of Field (DoF)

Aperture

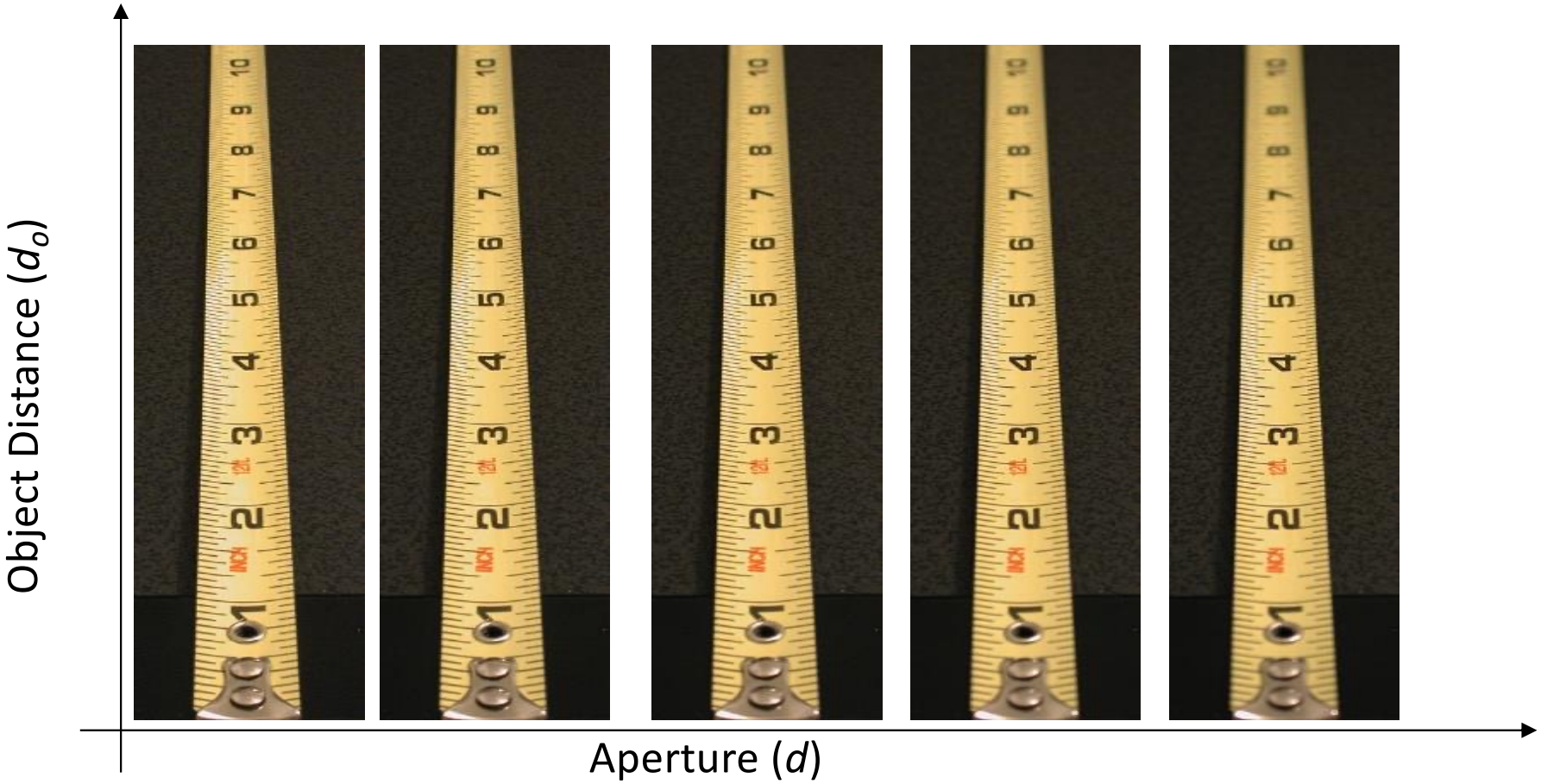


Focal Ratio = f / d

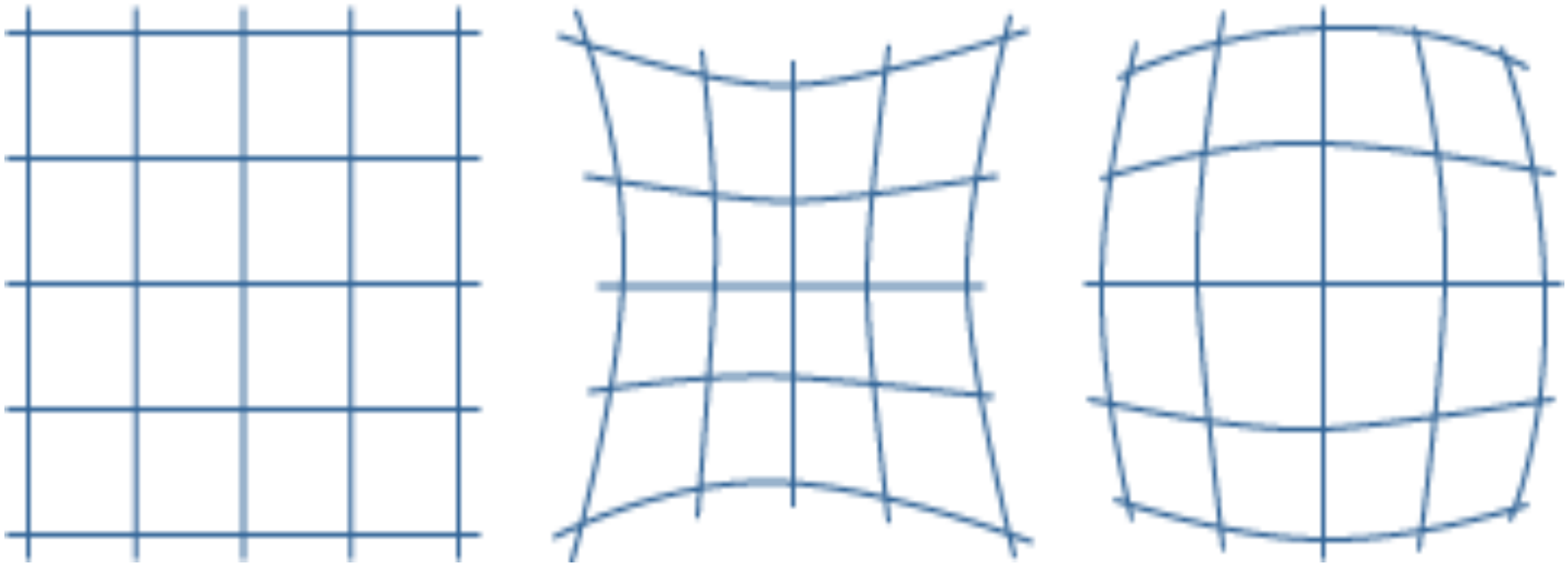
Aperture & DoF



Aperture & DoF



Geometric Distortion in Cameras



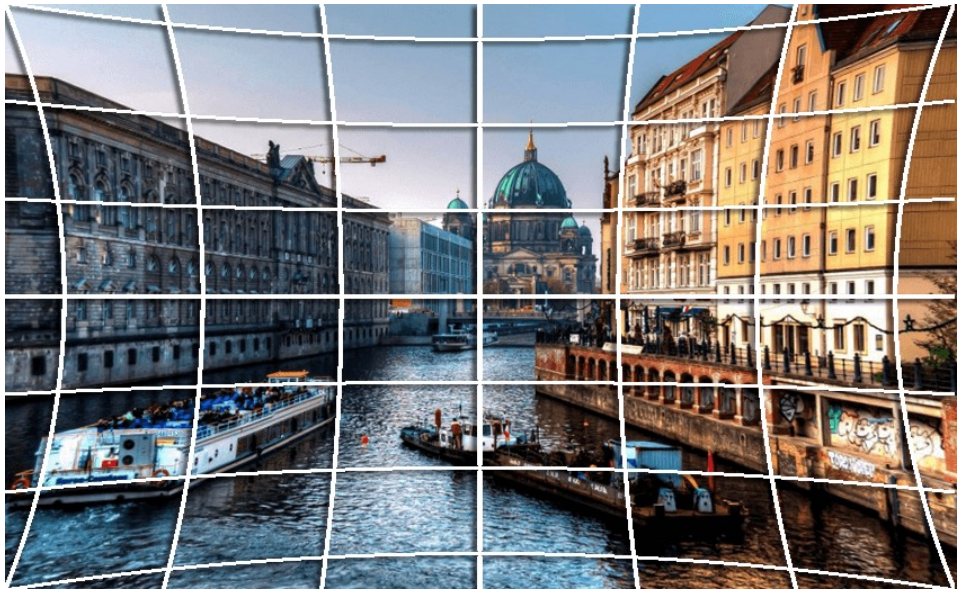
original pincushion barrel

Geometric Distortion in Cameras

pincushion

barrel

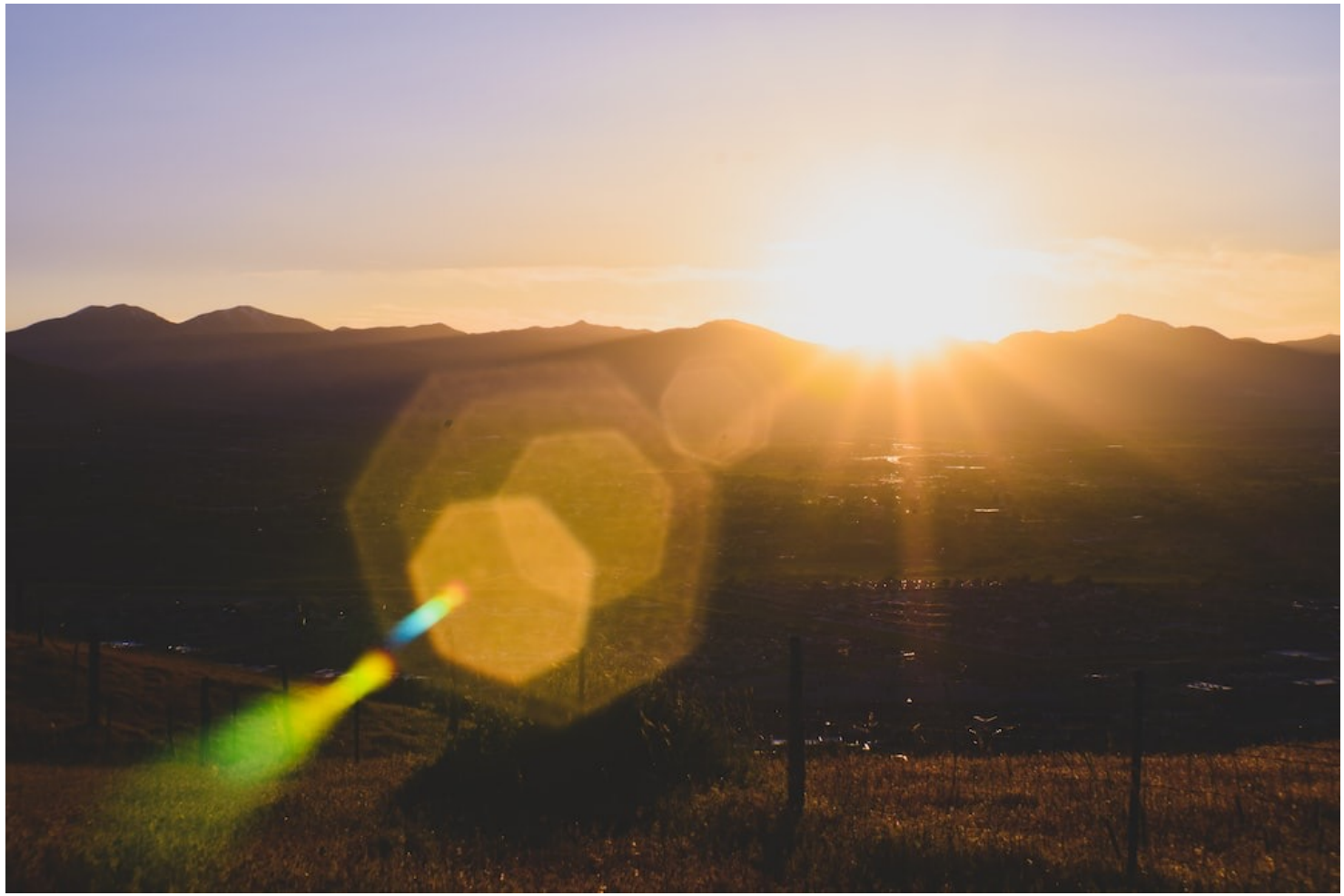
Geometric Distortion in Cameras



pincushion

barrel

Lens Flare



Chromatic Abrration

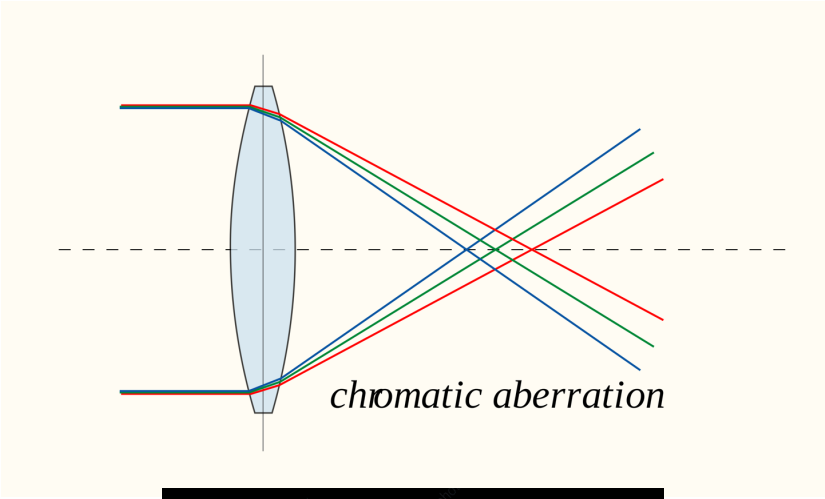


Image Sampling

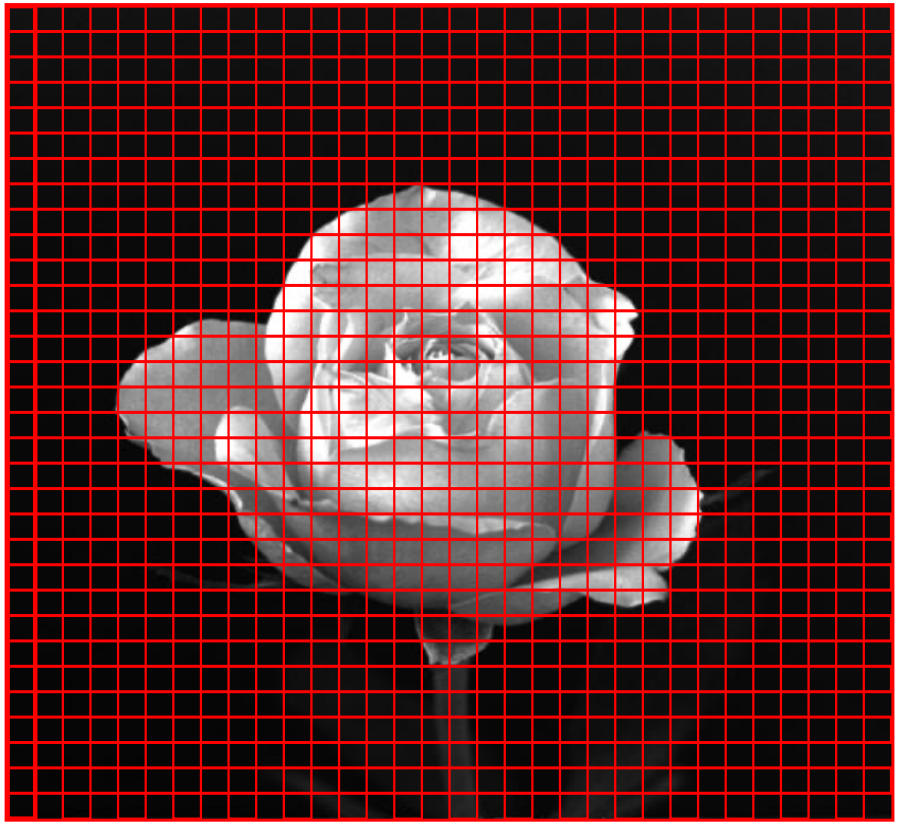


Image Sampling

- The number of samples in an image (number of sensor elements) is referred to as its resolution
- The resolution is typically represented as the product of number of samples in the horizontal and vertical directions in the image. e.g.: 32x32, 256x256, 640x480

Common Resolutions:

NTSC:	648 x 486
Typical Webcam:	1280 x 720
High-end SLR:	14204 x 10652
Hubbles Telescope*:	1,600 x 1,600

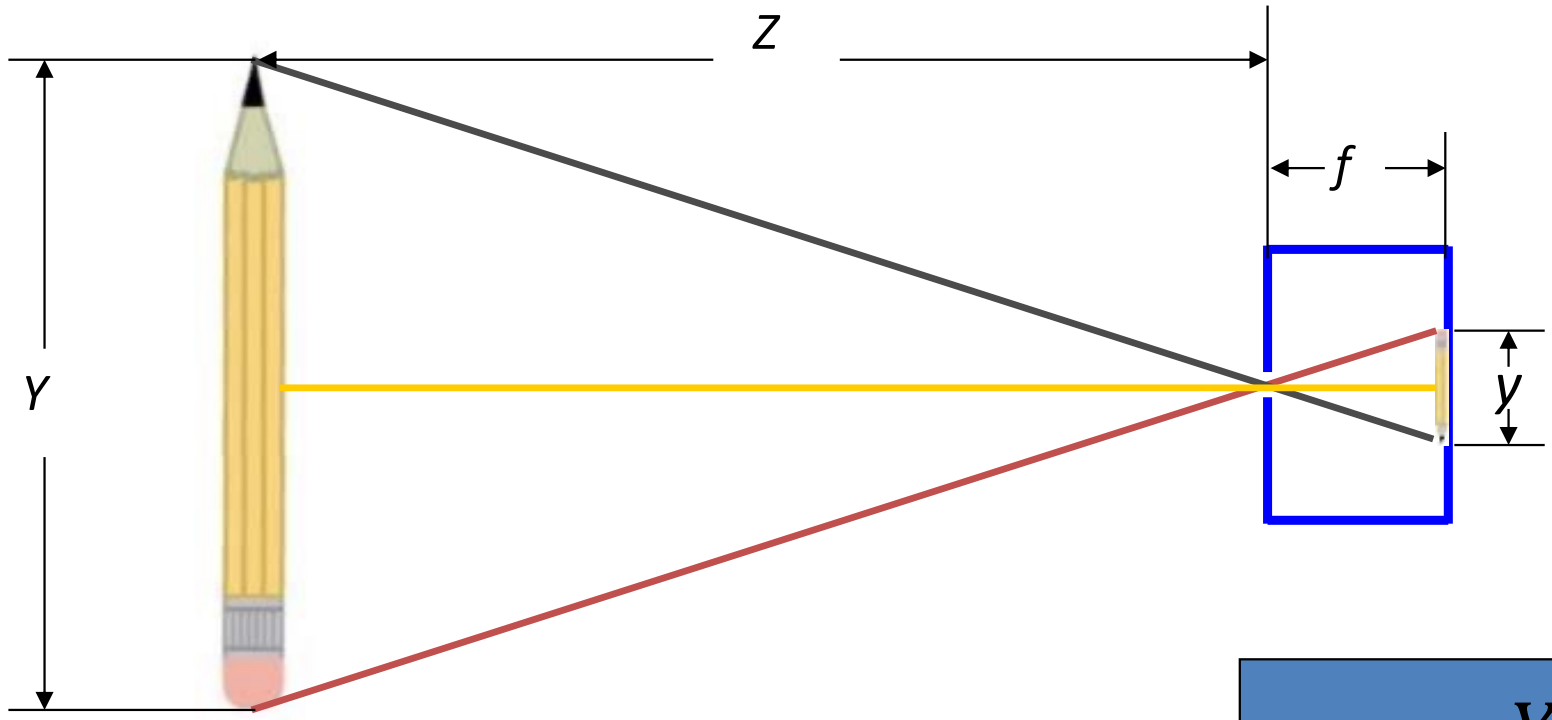
Objective of Camera Modeling

- Mathematically model what a camera does
  Also understand what the model means
- Getting the model for a real-world camera
  Estimation from real world measurements
- Special imaging configurations with simpler properties
  Simpler relationships
- General theory on fitting linear models under noisy observations
  Techniques that work across problems

Image Formation

- Form an image on the 2D image plane of the 3D world visible to it.
- Image is *behind* the lens; the scene is in front.
- 3D world is **projected** down to a 2D plane.
- Significant loss of information as one dimension is dropped.
- Mathematical depiction of this projection ...

We assume a Pin-hole Camera

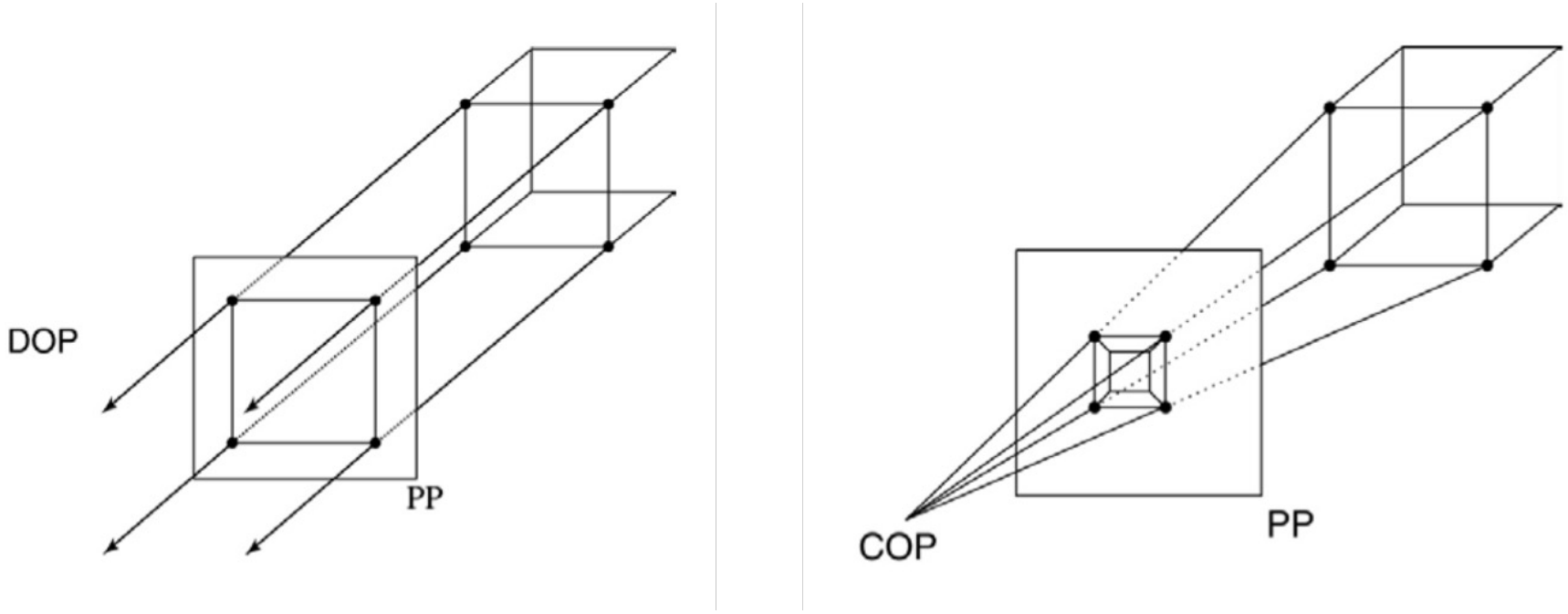


$$y = f\frac{Y}{Z}$$

Projecting 3D World to 2D Plane

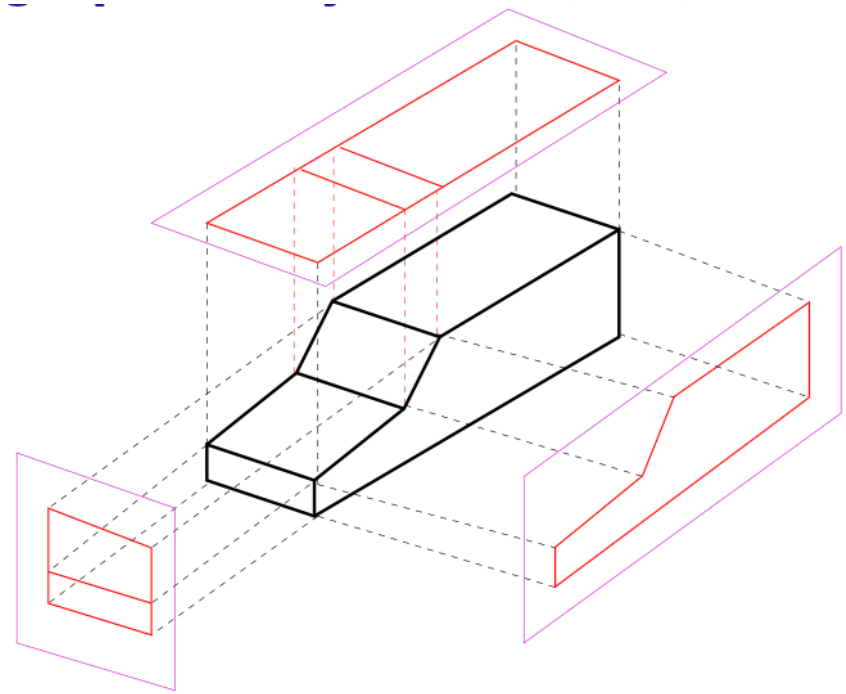
- Projection involves *projectors* starting from 3D points and hitting the 2D *projection plane*, forming the *image* of the point.
- Two types of projections.
- **Parallel projection**: Projectors are parallel to each other, all have the same *direction of projection* (**DOP**).
- **Perspective projection**: All projects pass through a point in space called the *centre of projection* (**COP**).

Projecting 3D World to 2D Plane



Parallel V/s Perspective Projection

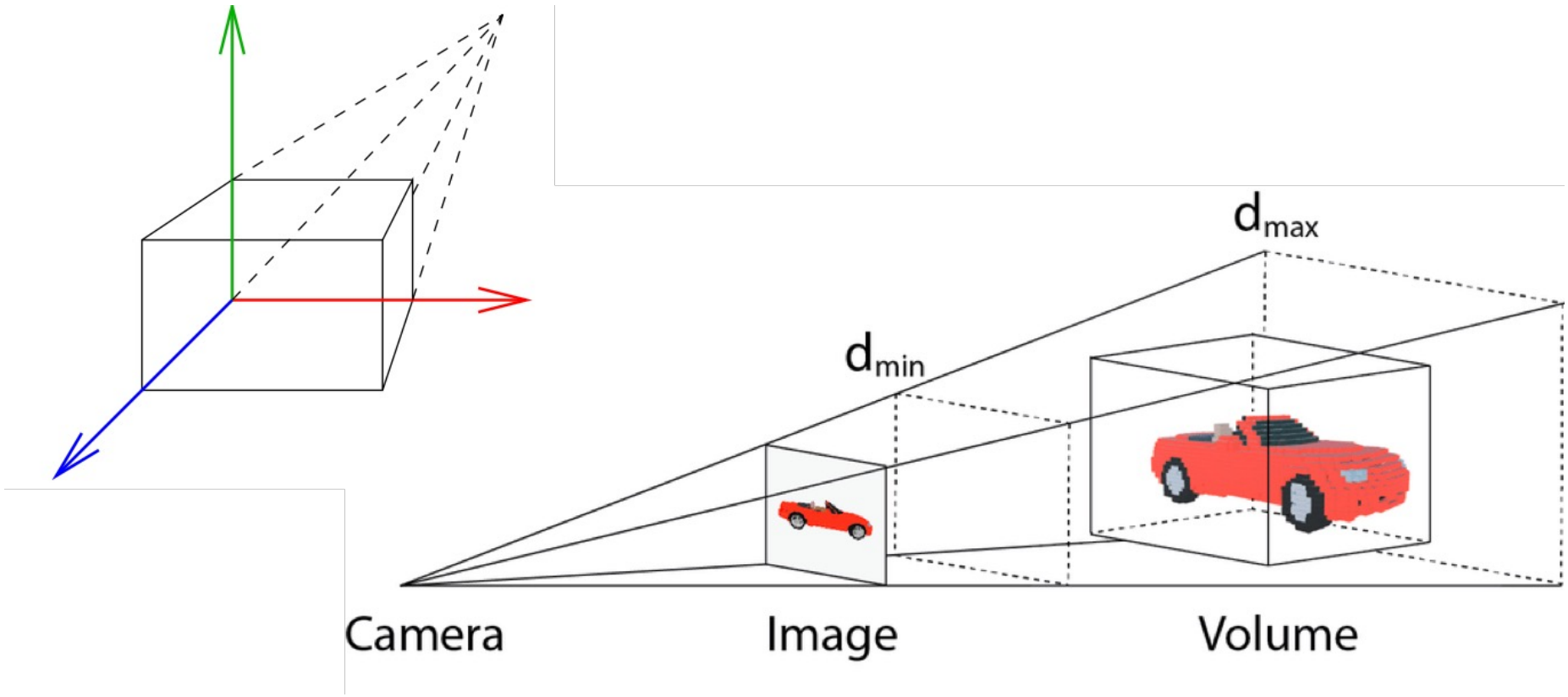
Orthographic (Parallel) Projection



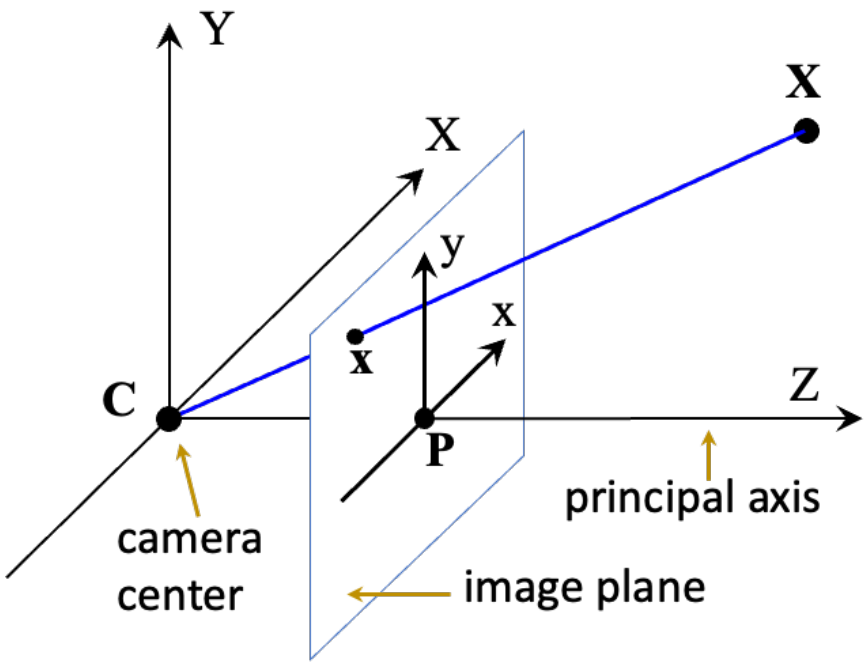
$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

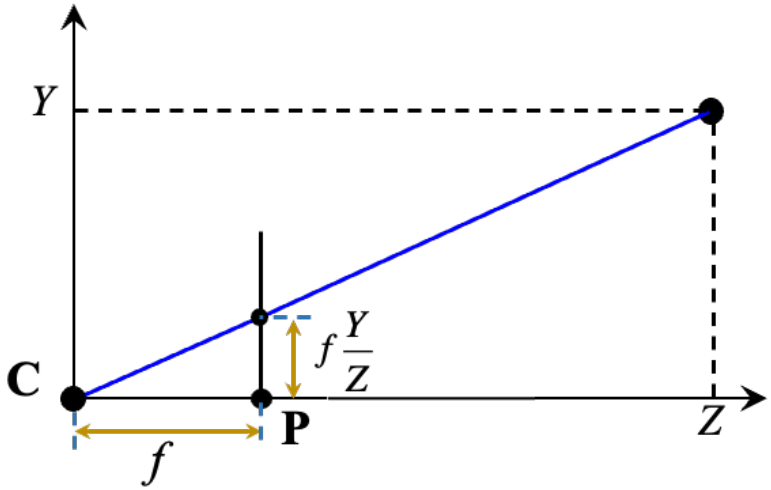
Projection matrix (homogeneous coordinates)

Perspective Projection



Geometry of Perspective Projection





$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

Cartesian image coordinates

Mathematical Model of Perspective Projection

$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{PX}$$

Projection matrix (homogeneous coordinates)

Geometry of Perspective Projection

$$x = f\frac{X}{Z}, \quad y = f\frac{Y}{Z}$$

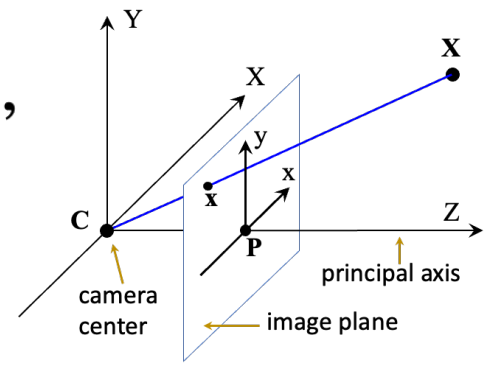
$$\mathbf{x} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \mathbf{PX}$$

Projection matrix (homogeneous coordinates)

Basic Camera Equation

A pinhole camera projects a 3D point $\mathbf{X_c}$ in camera coords to an image point $\mathbf{x}$ via the 3x4 **camera matrix** $\mathbf{P}$ as:

$$\mathbf{x} = \mathbf{P}\mathbf{X_c} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{I} \mid \mathbf{0}] \mathbf{X_c} = \mathbf{K}[\mathbf{I} \mid \mathbf{0}] \mathbf{X_c},$$



where $\mathbf{K}$ is the internal camera calibration matrix.

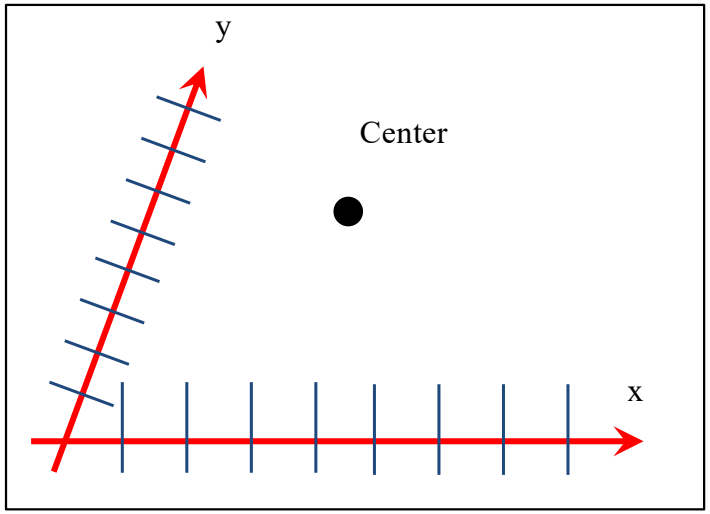
Note that:

- The camera is at the origin
- Z is the Camera or Optical axis
- Principal Point: Center of the image
- Focal length in pixel units
- Orthogonal image axes with uniform scale

Basic Camera Equation – A General Camera

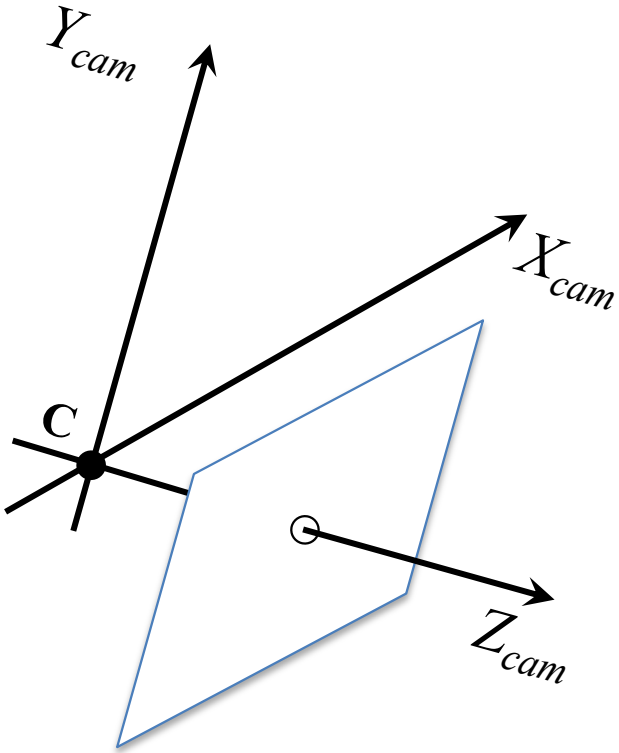
Image center at (x_0,y_0), Non-orthogonal axes with skew s, and different scales for axes with focal lengths, α_x and α_y.

$$\mathbf{K} = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

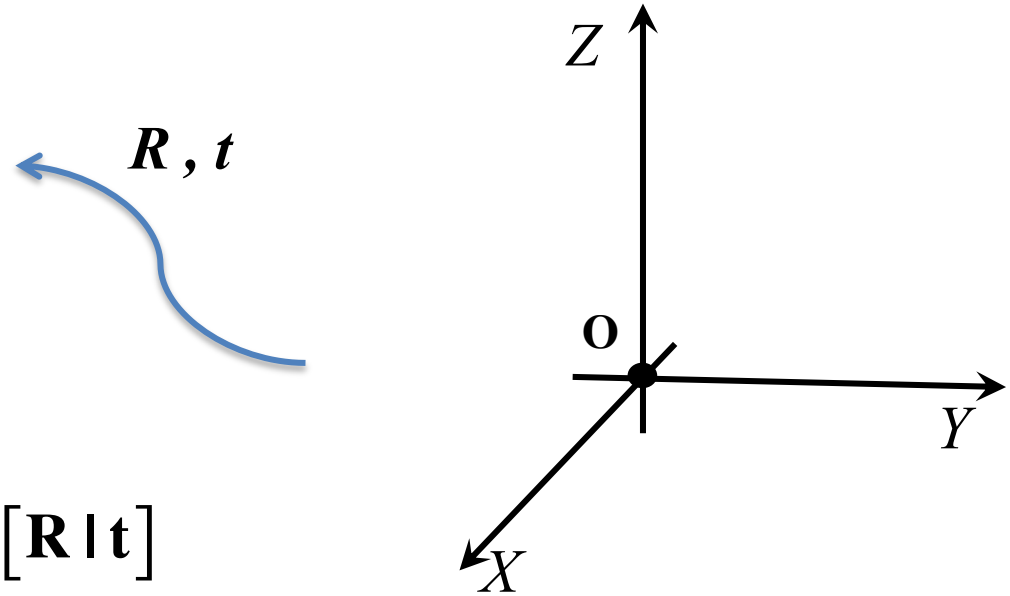


$\mathbf{K}$ an upper diagonal matrix with 5 degrees of freedom.

Basic Camera Equation – A General Camera



- General Setting: Camera is not at origin and Z is not the optical axis.
- Camera is at a point C in world coordinates. The camera axes are also rotated by a matrix R.



$$\mathbf{P} = \mathbf{K}\left[\mathbf{R} \mid \mathbf{t}\right]$$

Basic Camera Equation – A General Camera

- Camera and world are related by: $$\mathbf{X_c} = \begin{bmatrix} \mathbf{R} & \mathbf{-RC} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \mathbf{X_w}$$
- 2D projection $\mathbf{x}$ of a 3D point $\mathbf{X_w}$ given by:
  - $\mathbf{x} = \mathbf{K[\ I \mid 0\ ]\ X_c} = \mathbf{K\ [\ R \mid -RC\ ]\ X_w} = \mathbf{PX_w}$
- Camera matrix $\mathbf{P} = \mathbf{[KR \mid -KRC]} = \mathbf{[M \mid p_4]}$
- Common $\mathbf{K}$: $\begin{bmatrix} f & 0 & x_0 \\ 0 & f & y_0 \\ 0 & 0 & 1 \end{bmatrix}$ General $\mathbf{K}$: $\begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$

Thank You